Two robust approaches for classifying time series from dynamical systems

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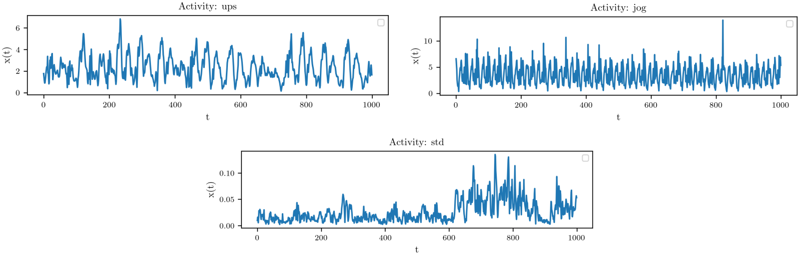
Problem

Time series classification originated from the K dynamical systems of the form

$$\begin{cases} \frac{d}{dt}\boldsymbol{z}(t) = f(\boldsymbol{z}), \\ \boldsymbol{z}(0) = \boldsymbol{z}_0, \end{cases} \tag{1}$$

assuming we have labeled sample series $\mathcal{D} = \{(\tilde{\boldsymbol{x}}_1(t_i), y_1), \dots, (\tilde{\boldsymbol{x}}_N(t_i), y_N)\}$.



Time series from different systems

"Direct"approach

1. Reconstruct diffeomorphic phase trajectories from the observed time series: Whitney or Takens embedding theorem.

$$\{\tilde{\boldsymbol{x}}(t_i)\} \to \{\boldsymbol{z}(t_i)\}$$

2. Learn vector fields of the systems $f_i(\boldsymbol{z})$ from the trajectories (NODE).

$$\hat{f} = \underset{f \in \mathcal{F}}{\arg\min}\ \mathcal{L}(\{\boldsymbol{z}(t_i)\}, f)$$

   Choose $\mathcal{L}, \mathcal{F}$ according to the prior knowledge of the system.

3. Associate a test trajectory with the system that gives least deviation from the *initial value problem* (IVP) (1).

$$k = \underset{k \in 1 \dots K}{\arg\min}\ \rho\big(\{\boldsymbol{z}(t_i)\}, \mathrm{IVP}(f_k, \boldsymbol{z}(0))\big)$$

Approach limitations

In steps 2-3 we need to compute IVP trajectories. For regular fields f and fixed $\boldsymbol{z}(0)$ IVP has unique solution. On practice, there are several factors that induce error in computing IVP$(f, \boldsymbol{z}(0))$.

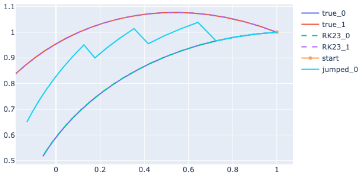
1. In many cases IVP has no analytic solution and *numerical solvers* are used. They induce numerical error and may have stability issues.
2. Learned field $\hat{f}$ from step 2 is an approximation of the real f. It contributes to the IVP error.
3. We have only noisy $\boldsymbol{z}(0)$ observation.
4. The dynamical system (1) may not be an adequate time series model globally. For example, the true field is not stationary $f(\boldsymbol{z}, t)$. But it can fairly approximates dynamics locally.

Trajectory segmenting

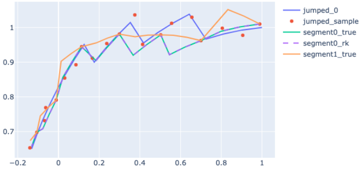
Do not rely for IVP($f, \boldsymbol{z}(0)$). We know full (noisy) trajectory $\{\boldsymbol{z}(t_i)\}$. Split it into parts and solve IVP per part. We can also add parameters for $\boldsymbol{z}(0)$ if noise is significant

$$\hat{f} = \underset{f \in \mathcal{F}, \{\boldsymbol{z}_0^s\}}{\arg\min} \sum_{s=0}^{S} \mathcal{L}(\{\boldsymbol{z}^s(t_i)\}, f, \boldsymbol{z}_0^s).$$

In terms of *bias-variance tradeoff* adding $\boldsymbol{z}_0$ parameters decrease bias but will increase variance. Trajectory segmenting also allows for *parallel* trajectory processing.



(a) No segmenting



(b) With segmenting

"Transition to model space"approach

Instead of analyzing trajectories let's analyze vector field f directly.

1. Split trajectory on the *segments* (with possibly adaptive sizes).
2. Assume linear field approximation is fair inside the segments

$$\dot{x} = f(\boldsymbol{x}) \approx f(\boldsymbol{x}_0) + \mathcal{I}\big|_{\boldsymbol{x}_0}(\boldsymbol{x} - \boldsymbol{x}_0)$$

   and learn it (i.e. $f(\boldsymbol{x}_0)$ and $\mathcal{I}\big|_{\boldsymbol{x}_0}$).
3. Do it for every train and test trajectory. In the end we have samples of f values across the phase space.

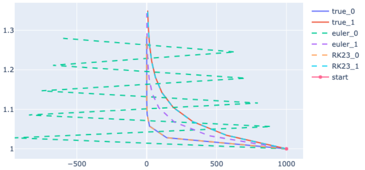
One possible solution is then:

1. Build global parametric approximation $\hat{f}_k,\ k \in 1 \ldots K$.
2. Classifier is the system with the minimum deviation of the vector field samples of the test trajectory

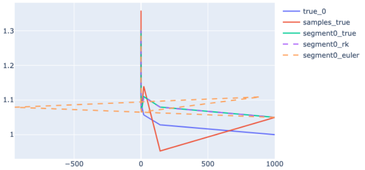
$$k = \underset{k \in 1 \ldots K}{\arg\min}\ \rho\big(\{f(t_i)\}, \{\hat{f}_k(t_i)\}\big)$$

Experiments to justify trajectory segmenting

Trajectory segmenting helps to stabilize numerical solvers.

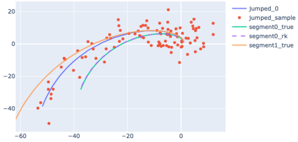


(c) No segmenting

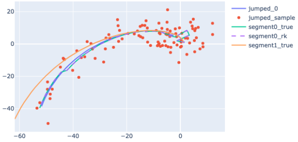


(d) With segmenting

Trajectory segmenting is robust under noisy samples.



(e) No segmenting



(f) With segmenting

Future work

1. Adaptive segmenting.
2. Relation to non-stationary systems and stochastic differential equations.
3. More deep limitations analysis.